

UNIT 5

WORKING WITH EXCEL



Learning Objectives

- Create and manage multiple sheets in Excel, including adding, renaming, and organizing them within a workbook.
- Insert, delete, and adjust rows and columns in Excel to manage data effectively.
- Utilize Excel's Find and Select feature to search for specific data within a given range.
- Apply formatting to rows and columns by adjusting sizes, freezing panes, and using gridlines.
- Copy, move, and organize sheets within a workbook to enhance data organization.
- Use basic Excel functions (e.g., SUM) to perform calculations across multiple sheets.

Working with Sheets

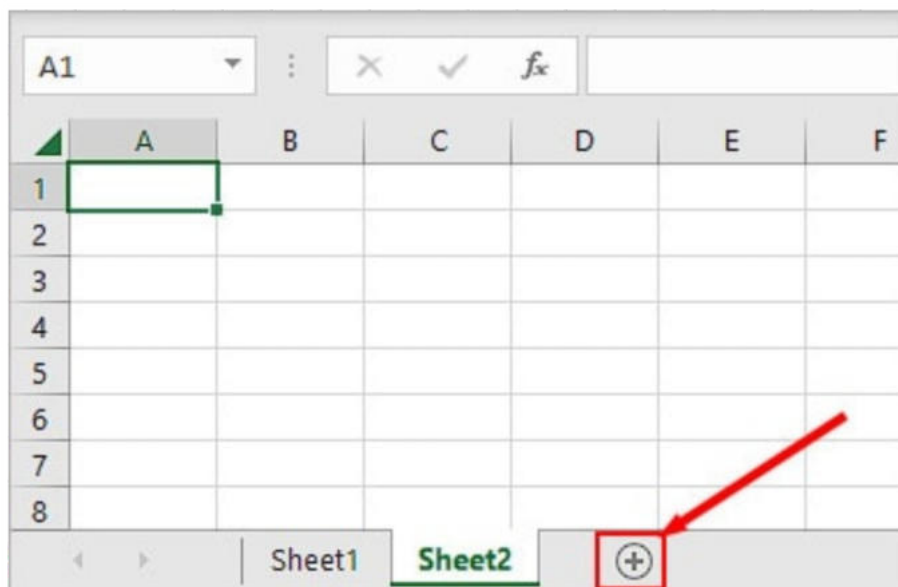
Microsoft Excel is a powerful tool for organizing and analyzing data. In this chapter, you will learn how to work with different sheets, perform common tasks, and explore various tools available in Excel. Let's start with working on sheets in Excel!

1. Creating New Sheets

In Excel, you can create a new sheet to organize your data. Each workbook can contain multiple sheets.

To create a new sheet:

1. Open Excel.
2. At the bottom of your screen, you will see a tab labelled "Sheet1" (or any existing sheet name).
3. Click the "+" button next to the sheet tabs to add a new sheet.

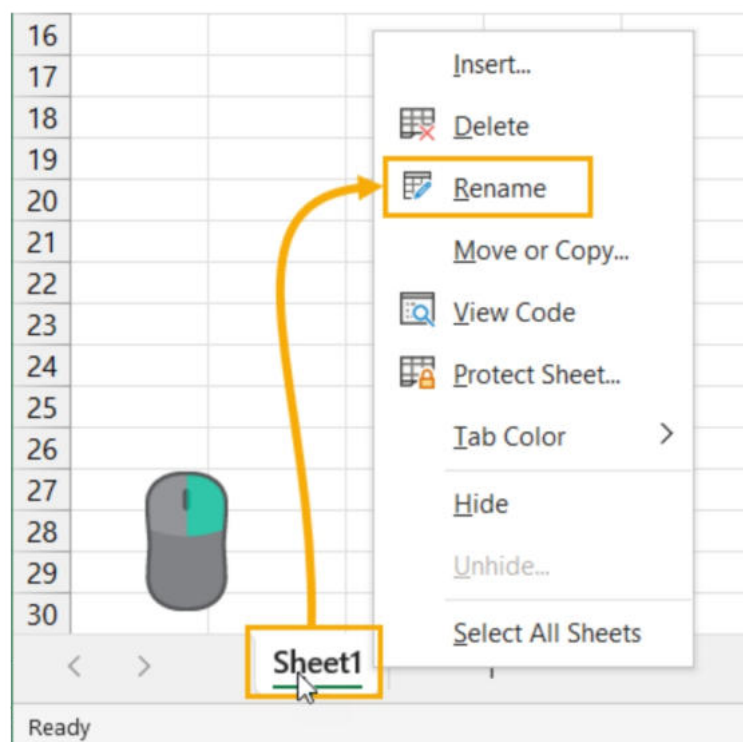


2. Renaming Sheets

Renaming sheets helps you organize your work by giving each sheet a relevant name.

To rename a sheet:

1. Right-click on the sheet tab (e.g., Sheet1).
2. Select Rename from the menu.
3. Type the new name for your sheet and press Enter.

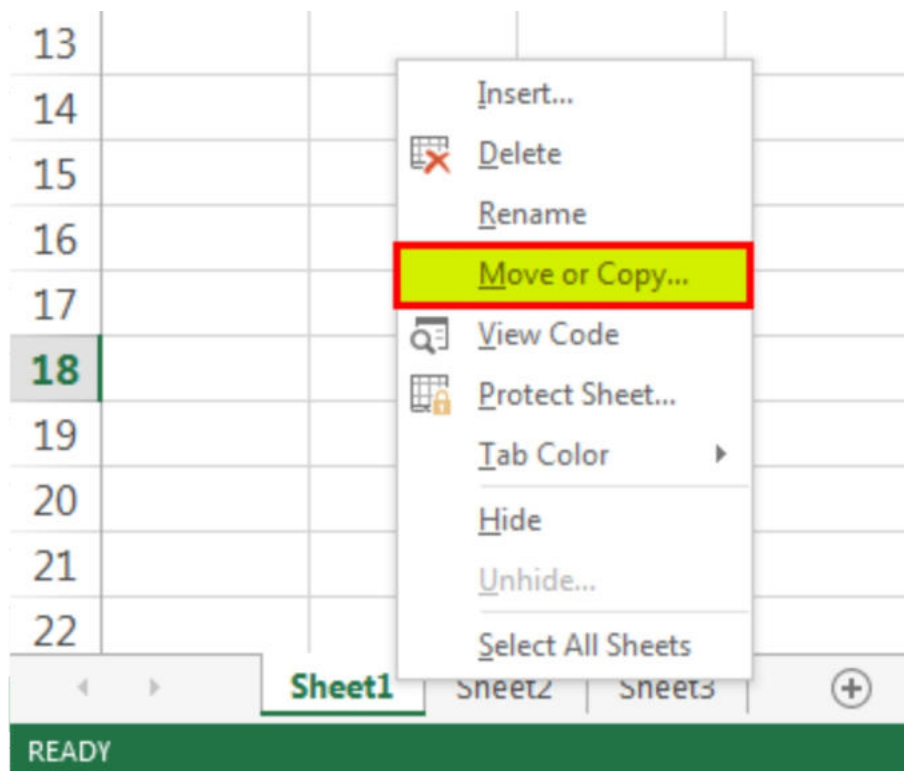


3. Copying / Moving Sheets

You can copy or move sheets within the same workbook or between different workbooks.

To copy or move a sheet:

1. Right-click on the sheet tab you want to copy or move.
2. Select Move or Copy.
3. Choose where you want to move the sheet, or check the Create a copy box to copy it.
4. Click OK.



Inserting New Row and Column

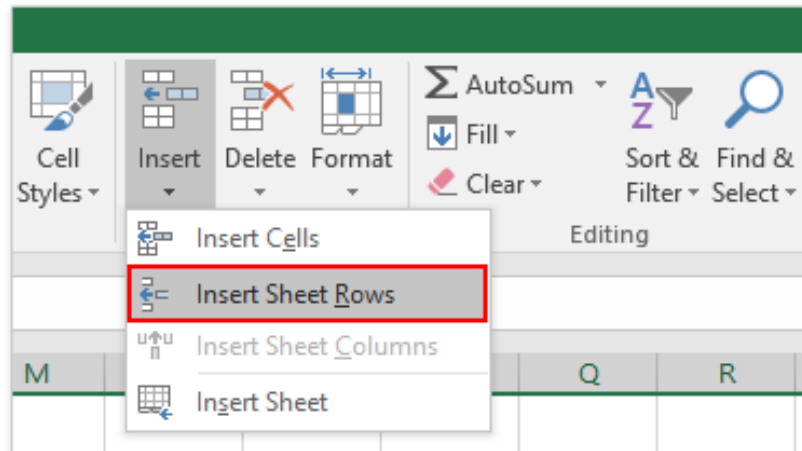
You can add new rows and columns in Excel to expand your data table.

1. To insert a new row:

1. Right-click on the row number where you want the new row to appear.
2. Click Insert from the menu.

2. To insert a new column:

1. Right-click on the column letter where you want the new column to appear.
2. Click Insert.



Deleting / Hiding Rows and Columns

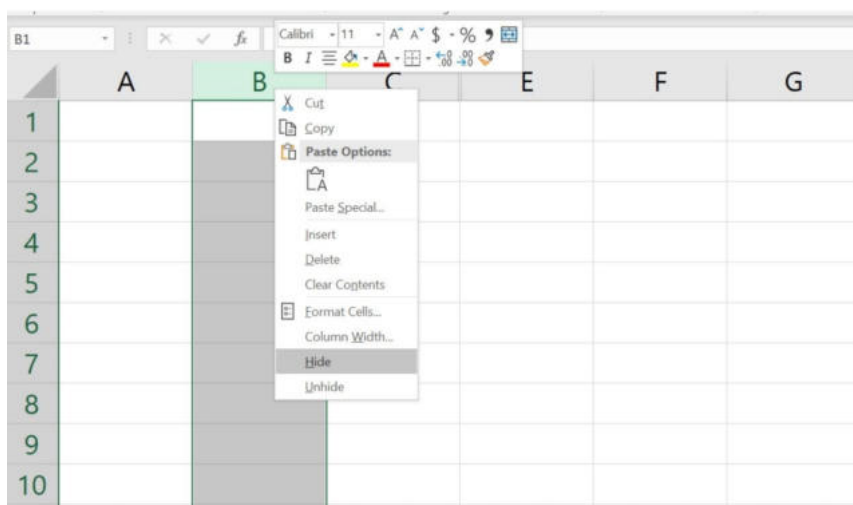
You can delete rows or columns if you no longer need them, or hide them to temporarily remove them from view.

1. To delete a row or column:

1. Right-click on the row number or column letter you want to delete.
2. Click Delete.

2. To hide a row or column:

1. Right-click on the row or column.
2. Click Hide.



Changing the Width and Height of Rows and Columns

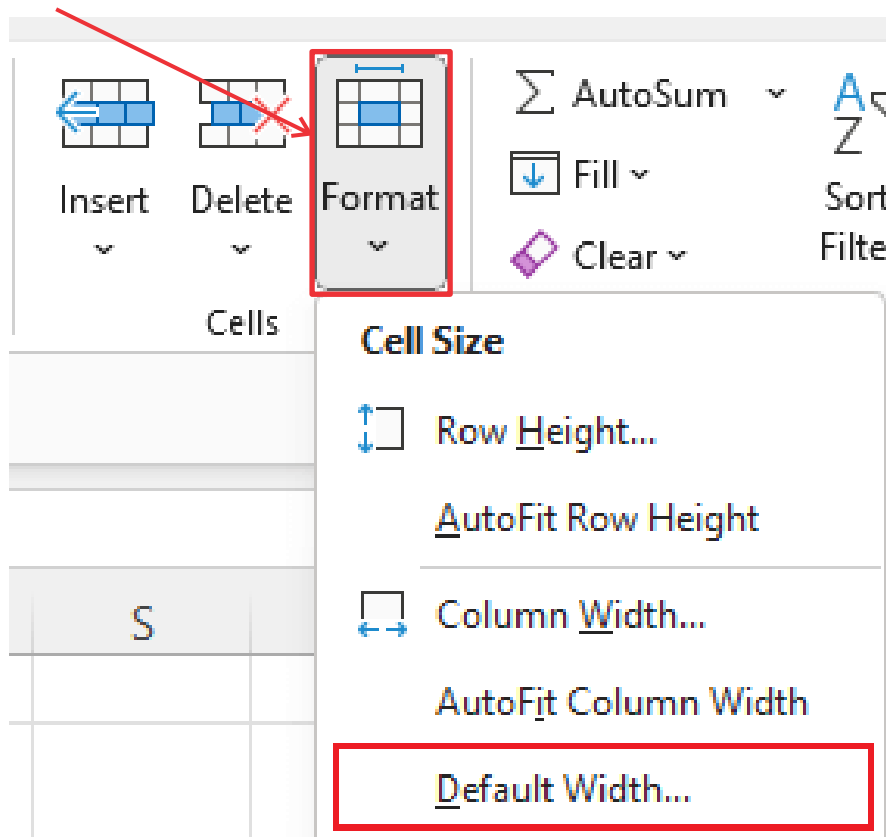
Sometimes, you may need to adjust the size of rows or columns to fit your data.

1. To change the width of a column:

1. Hover your mouse over the line between two column letters.
2. When the cursor changes to a double arrow, click and drag to adjust the width.

2. To change the height of a row:

1. Hover your mouse over the line between two row numbers.
2. When the cursor changes to a double arrow, click and drag to adjust the height.

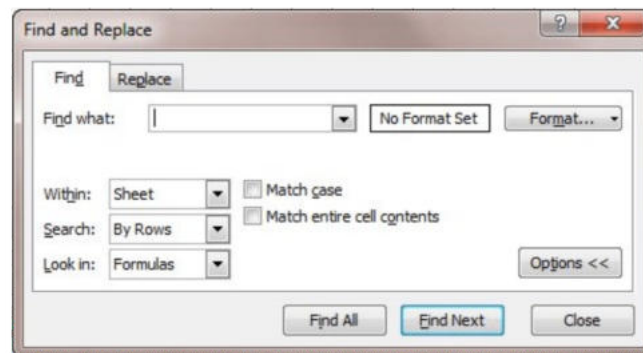


Find and Select Data in a Given Range

Excel allows you to quickly find and select data in a specific range.

To find data:

1. Press Ctrl + F (Windows) or Cmd + F (Mac).
2. Type the data you want to search for in the "Find" box.
3. Click Find All or Find Next.



View Menu

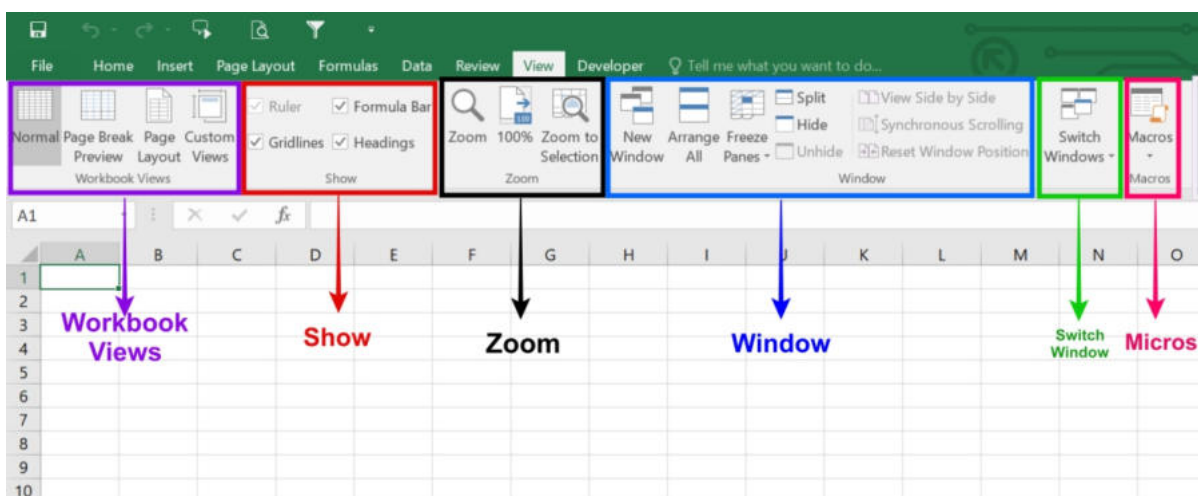
The **View** menu in Excel offers several options to improve your workspace. Let's look at some key options.

1. Gridlines

Gridlines help you see the boundaries of cells. You can toggle them on or off.

To show/hide gridlines:

1. Go to the **View** tab.
2. Check or uncheck the **Gridlines** option under the "Show" group.

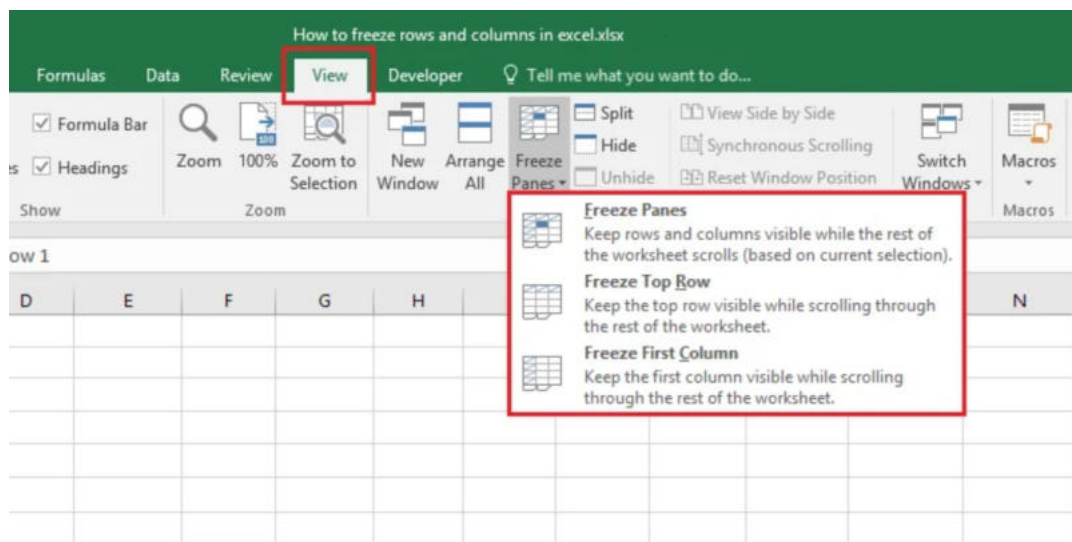


2. Freeze Panes

The Freeze Panes option allows you to lock certain rows or columns in place so that they stay visible as you scroll through your worksheet.

To freeze panes:

1. Select the row or column where you want to freeze.
2. Go to the View tab and click on Freeze Panes.
3. Choose the desired option (e.g., Freeze Top Row, Freeze First Column).

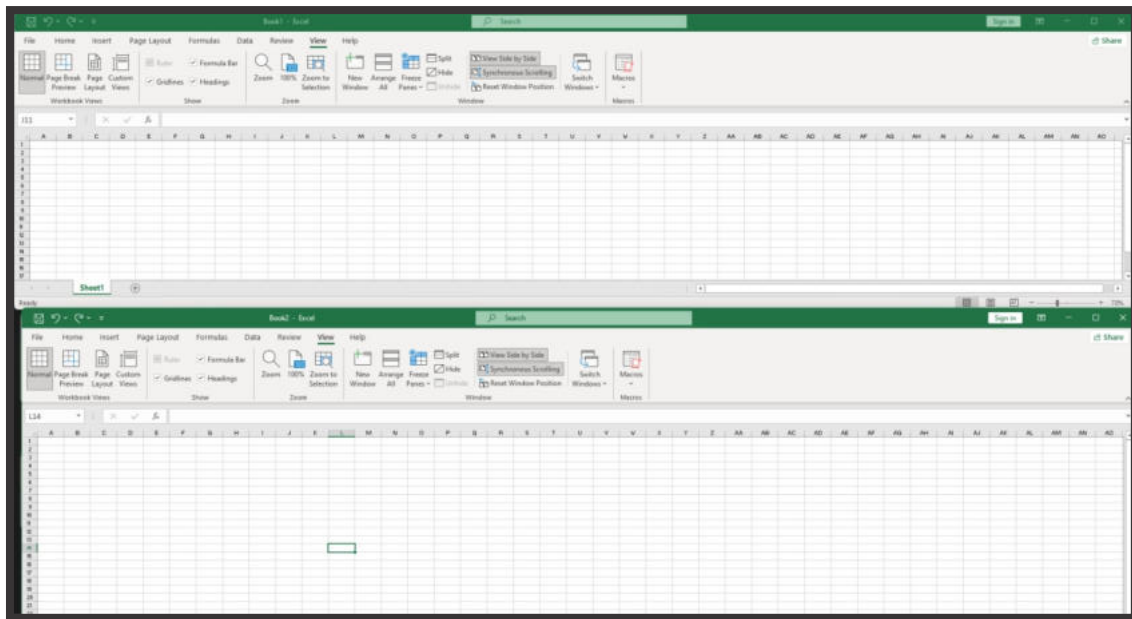


View Side by Side

The View Side by Side feature allows you to compare two worksheets at once

To view side by side:

1. Open two workbooks in Excel.
2. Go to the View tab and click View Side by Side.



KNOW THIS?

MS Excel (2016, 2019, 2021, and Microsoft 365) maintain a consistent layout, with updates introducing new charts, dynamic arrays, advanced functions (like XLOOKUP), and AI-driven data analysis tools in Microsoft 365.

EXERCISES

01: Choose the correct answer.

1. What is the shortcut to create a new sheet in Excel?

- (a) Ctrl + N
- (b) Ctrl + Shift + N
- (c) Click on the "+" icon
- (d) None of the above

2. Which option is used to freeze the top row in Excel?

- (a) View Side by Side
- (b) Freeze Panes
- (c) Gridlines
- (d) None of the above

3. What happens when you click the "Delete" option after right-clicking a row or column?

- (a) It hides the row/column
- (b) It removes the row/column permanently
- (c) It moves the row/column to another location
- (d) It copies the row/column

4. Which feature allows you to search for a specific word or number in Excel?

- (a) View Side by Side
- (b) Find and Select
- (c) Freeze Panes
- (d) Insert Row

5. To insert a new column in Excel, you should:

- (a) Right-click on the column letter
- (b) Press Ctrl + C
- (c) Press Shift + A
- (d) None of the above

6. Which feature lets you view multiple worksheets side by side?

- (a) Freeze Panes
- (b) View Side by Side
- (c) Gridlines
- (d) None of the above

7. What is the purpose of renaming a sheet in Excel?

- (a) To delete the sheet
- (b) To organize sheets by giving them relevant names
- (c) To move the sheet
- (d) To apply formatting to the sheet

8. How can you adjust the width of a column in Excel?

- (a) Right-click on the column letter and choose Adjust Width
- (b) Hover your mouse over the line between two column letters and drag
- (c) Use the View tab and click Column Width
- (d) Double-click anywhere on the column

9. What does the Gridlines option in the View menu do?

- (a) Deletes rows and columns
- (b) Toggles the visibility of cell boundaries
- (c) Changes the width of columns
- (d) Highlights specific cells

10. Which feature allows you to copy a sheet to another workbook?

- (a) Freeze Panes
- (b) Find and Select
- (c) Move or Copy
- (d) View Side by Side

02: Fill in the blanks.

1. To insert a new row, you right-click on the _____ and select "Insert."
2. To adjust the width of a column, you drag the line between the _____ letters.
3. To show or hide gridlines in Excel, go to the _____ tab.
4. When you right-click a sheet tab, the options to "Rename," "Move," or "_____" will appear.
5. If you want to lock the top row in place, you can use the _____ feature under the View tab.
6. If you want to search for data, press the shortcut _____.

03: Tick True or False for each statement.

	TRUE	FALSE
You can have multiple sheets in a single workbook.	☐	☐
The "Find" feature in Excel allows you to search for a specific number or text in your data.	☐	☐
You can freeze more than one row or column using the Freeze Panes feature.	☐	☐
Gridlines in Excel are visible by default.	☐	☐
Right-clicking a column letter gives the option to "Delete" that column.	☐	☐
The "View Side by Side" option can be used to compare two different Excel files.	☐	☐

04: Match the column

A
1. Freezing Panes
2. Insert Row
3. Delete Column
4. Gridlines

B
a) Add rows
b) Lock rows or columns in place
c) Expand or contract columns
d) Remove columns

05: Answer the following questions briefly.

1. How can you insert a new sheet in Excel?
2. What is the purpose of the "Find" tool in Excel?
3. Describe how to change the size of a row in Excel.
4. How do you move or copy a sheet in Excel?
5. What is the difference between freezing a row and hiding it in Excel?
6. How can you adjust the width of a column?

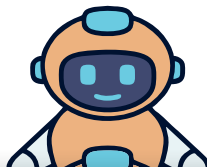
06: Provide detailed answers for the following questions.

1. Explain the steps for inserting and deleting rows and columns in Excel. How would you handle situations where you need to adjust the size of rows and columns to fit the data? Include an example of when you might need to hide a row or column.
2. In what situations would you use the Freeze Panes feature in Excel? Explain how freezing rows and columns can help you work more efficiently when managing large spreadsheets.
3. What is the importance of gridlines in Excel? Describe how you would show and hide gridlines in a worksheet and why this option might be useful when preparing data for presentation or printing.



Excel

Lab Activity 1: 1. Create and Organize a Workbook

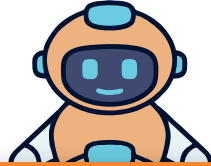


To practise creating, renaming, and organizing sheets in Excel.

Instructions:

1. Open a new Excel workbook.
2. Create at least three sheets and name them according to different topics (e.g., "Maths", "Science", "English").
3. Rename each sheet to reflect its content (e.g., "Sheet-1" to "Math Homework").
4. Move one sheet to a new position in the workbook.
5. Save the workbook for future use.

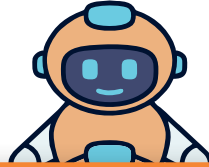
Lab Activity 2: Resize and Organize Data



To practise resizing rows and columns
and organizing data.

Instructions:

1. Create a table with 5 columns and 10 rows.
2. Add some data in the cells, such as names, numbers, and dates.
Resize the columns to fit the data (e.g., adjust the column width to fit the longest entry).
4. Change the height of some rows to fit longer text entries.
5. Adjust the table layout to make it neat and easy to read.

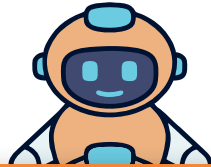
Lab Activity 3: Using Freeze Panes and Gridlines

To explore the use of Freeze Panes and Gridlines to manage large datasets.

Instructions:

1. Open a large dataset in Excel (e.g., 50 rows and 10 columns).
2. Freeze the top row so that it stays visible when you scroll down.
3. Turn on the gridlines to help you see the boundaries of each cell.
4. Scroll through the data to ensure the top row stays frozen and visible.

Lab Activity 4: Find and Select Data



To practise using the "Find" tool to locate specific data within a worksheet.

Instructions:

1. Open a dataset in Excel with at least 100 entries.
2. Use the "Find" tool (Ctrl + F) to locate a specific value (e.g., a student's name or a particular number).
3. Use the "Find All" option to find all occurrences of a value within the dataset.
4. Practise selecting the found data and marking it with a specific colour to make it stand out.

Lab Activity 5: Creating a Budget Sheet in Excel

Students will apply their knowledge of Excel by creating a budget sheet that includes multiple sheets, rows, columns, formatting, and other tools they have learned. They will practise tasks such as inserting, deleting, and adjusting rows/columns, working with multiple sheets, and freezing panes.

Instructions:**Step 1: Set Up Your Workbook**

1. Open Microsoft Excel.
2. Create a new workbook.
3. Add two sheets:
 - Rename the first sheet to "Income".
 - Rename the second sheet to "Expenses".
4. Save the workbook as "Monthly Budget.xlsx".

Step 2: Enter Data in the "Income" Sheet

1. In the "Income" sheet, enter the following columns:

- **A1:** "Source of Income"
- **B1:** "Amount"

2. Under the "Source of Income" column, enter:

- **A2:** "Salary"
- **A3:** "Freelance Work"
- **A4:** "Investments"

3. In the "Amount" column, enter:

- **B2:** "3000"
- **B3:** "500"
- **B4:** "200"

4. **Apply formatting:**

- Bold the headers in **A1** and **B1**.
- Use the "Currency" format for the values in column B.

Step 3: Enter Data in the "Expenses" Sheet

1. Switch to the "Expenses" sheet.

2. Enter the following columns:

- **A1:** "Expense Category"
- **B1:** "Amount"

3. Under the "Expense Category" column, enter:

- **A2:** "Rent"
- **A3:** "Utilities"
- **A4:** "Groceries"
- **A5:** "Entertainment"

4. In the "Amount" column, enter:

- **B2:** "1000"
- **B3:** "150"
- **B4:** "300"
- **B5:** "100"

5. **Apply formatting:**

- Bold the headers in **A1** and **B1**.
- Use the "Currency" format for the values in column B.

Step 4: Add Totals

1. In the "Income" sheet, below your income data, create a total row:

- In **A5**, type "Total Income".
- In **B5**, use the **SUM** function to calculate the total of all income (e.g., =SUM(B2:B4)).

2. In the "Expenses" sheet, below your expenses data, create a total row:

- In **A6**, type "Total Expenses".
- In **B6**, use the **SUM** function to calculate the total of all expenses (e.g., =SUM(B2:B5)).

Step 5: Calculate Remaining Budget

1. In the "Income" sheet, in **A6**, type "Remaining Budget".

2. In **B6**, calculate the remaining budget by subtracting total expenses from total income (e.g., =B5-Expenses!B6).

Step 6: Adjust Rows and Columns

1. Adjust the width of columns A and B to fit the data properly.
2. Insert a new row between the "Total Income" and "Remaining Budget" if necessary, to improve readability.

3. Freeze Panes:

- Freeze the top row in both sheets to keep the column headers visible while scrolling. To do this, go to the View tab, click "Freeze Panes", and select "Freeze Top Row".

Step 7: Optional Enhancements

1. **Hide a row** (e.g., to hide a row that's no longer needed).
2. **Insert a new column** for "Date" to track when each income and expense occurred.
3. Use conditional formatting to highlight expenses that exceed a certain value (e.g., anything over \$200 is highlighted in red).

Step 8: Save and Submit

1. Review your work and make sure everything is correctly formatted and calculations are accurate.
2. Save the file.
3. Submit the workbook as your lab activity.